CLAUDINE HANDALI

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TECHNICAL SKILLS

Programming: Python, SQL,Typescript, Java, Javascript, HTML/CSS
Skills: Real-time telemetry, Time Series Analysis, Data Modeling, Agentic AI
Tools: AWS, Git, DuckDB, Pandas, sckit-learn, D3.js, Mapbox GL JS, Flask, React, PyTorch, XGBoost

EDUCATION

University of California, San Diego La Jolla, CA
Bachelor of Science (B.S.) in Data Science | Major GPA: 3.9 Expected June 2026
Relevant Courseworks: Recommender Systems and Web mining, Scalable Systems and Analytics, Data Visualizations, Representation Learning, Data Analysis, Data Management, Data Structures and Algorithms, Machine Learning

WORK EXPERIENCE

Lead Student Researcher January 2026 - Present
University of California, San Diego Center for Energy Research La Jolla, CA
- Lead and mentor a team of 4 students in developing an interactive data dashboard analyzing the impact of EV adoption on oil import reduction across 25+ countries.
- Built quantitative models processing 10,000+ data points (energy consumption, trade flows, vehicle adoption rates) to estimate potential reductions in oil import expenditures.
- Presented findings to 10 faculty and researchers; dashboard adopted as a teaching and research tool within the center.

Student Researcher June 2025 - October 2025
University of California, San Diego La Jolla, CA
- Analyzed U.S. computing electricity consumption using 9TB of Intel DCA telemetry to profile CPU power-state residency and burst behavior, delivering plots and a technical brief that informed the team's next priorities.
- Built data processing pipelines and statistical models to evaluate computing efficiency across workloads.
- Standardized analysis notebooks using DuckDB on partitioned Parquet files.

AI Developer Intern March 2025 - July 2025
Service Agent La Jolla, CA
- Designed and deployed AI-driven automation agents, including a GPT-4-powered cold-calling system, using Python, Twilio, n8n, and SupaBase.
- Developed data enrichment (web scraping + APIs) for CRM systems, improving data quality and completeness by 50%.
- Built predictive outreach algorithms integrating AI dialogue management, boosting conversion rates by 40%.

PROJECT EXPERIENCE

Predicting Recipe Popularity via Ratings
Python, Statsmodels, Multiple Linear Regression
- Analyzed 1.1M+ recipe interactions to identify key drivers of user ratings; found simpler recipes (≤8 ingredients) received 12–18% higher average ratings.
- Built a regression model to predict recipe popularity ($R^2 = 0.42$), quantifying the impact of complexity and prep time on user preferences.
- Improved model reliability through multicollinearity and heteroskedasticity checks.

End-to-End Analysis of U.S. Power Outage Durations
Scikit-learn, Pandas, Random Forest Regression
- Modeled 100K+ outage events using Random Forest regression, reducing RMSE by 28% vs. linear baseline.
- Improved predictive performance from $R^2 = 0.51$ to 0.68 through hyperparameter tuning and feature engineering.
- Identified a 22% prediction error gap across outage types, highlighting bias and informing mitigation strategies

Interactive Climate Data System
JavaScript, D3.js, Data Storytelling
- Processed and cleaned 170+ years (1850–2020) of NOAA precipitation data spanning 3,000+ countries and 60K+ data points.
- Built an interactive D3.js climate dashboard and emissions simulator, increasing user engagement 3× compared to static reports
- Designed intuitive visualizations enabling non-technical users to explore climate patterns without data expertise.